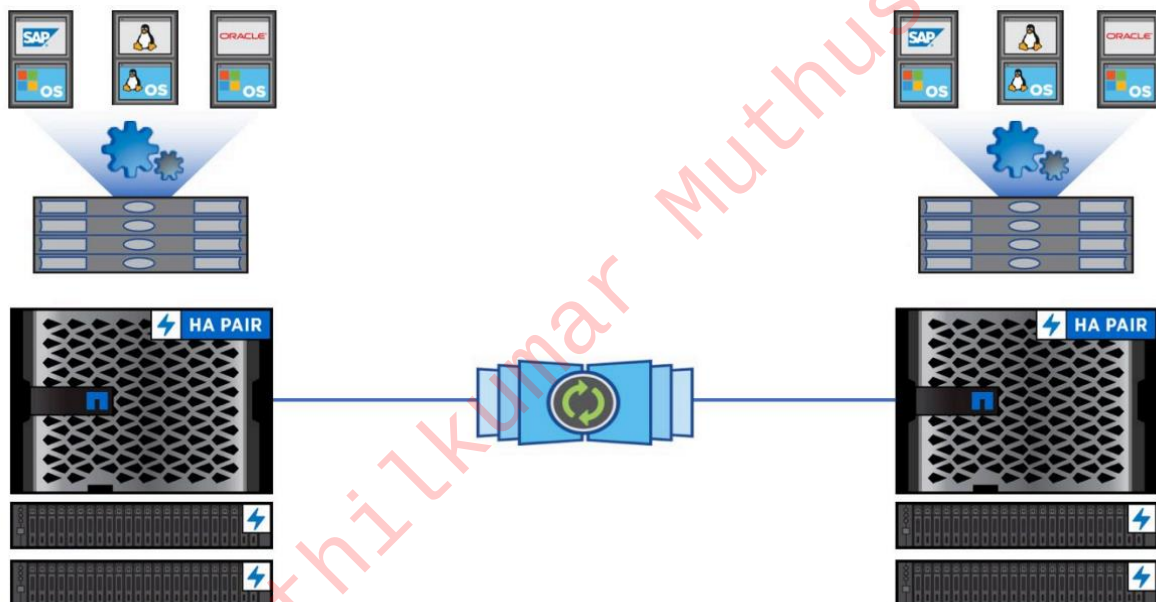


NetApp ONTAP Metrocluster Configuration

Overview

NetApp® MetroCluster™ configurations combine array-based clustering with synchronous replication to deliver continuous availability, immediately duplicating all of your mission-critical data on a transaction-by-transaction basis. MetroCluster configurations enhance the built-in high availability and nondisruptive operations of NetApp hardware and ONTAP® storage software, providing an additional layer of protection for the entire storage and host environment.



With MetroCluster, you can:

- Protect against hardware, network, or site failure with a transparent switchover.
- Eliminate planned and unplanned downtime and change management.
- Upgrade hardware and software without disrupting operations.
- Deploy without complex scripting, application, or operating system dependencies.

- Achieve continuous availability for VMware, Microsoft, Oracle, SAP, or any critical application.

NetApp MetroCluster enhances the built-in high-availability (HA) and nondisruptive operations of NetApp hardware and ONTAP storage software, providing an additional layer of protection for the entire storage and host environment. Whether your environment is composed of standalone servers, HA server clusters, or virtualized servers, MetroCluster seamlessly maintains application availability in the face of a site storage outage.

Such an outage could result from loss of power, cooling, or network connectivity; a storage array shutdown; or operational error. MetroCluster is an array-based, active-active clustered solution that eliminates the need for complex failover scripts, server reboots, or application restarts.

MetroCluster provides the following benefits:

- Multiprotocol support for a wide range of SAN and NAS client and host protocols.
- Synchronous replication supported over FC or IP networks.
- No charge for MetroCluster functionality.
- Mirror only critical data: support for mirrored and unmirrored aggregates.
- Easy import of third party storage with Foreign LUN Import (FLI).
- Storage and network efficiencies achieved from deduplication, compression, and compaction.

- Integration with NetApp SnapMirror® technology to support asynchronous replication, distance, and SLA requirements.

Data protection with NetApp SyncMirror technology

At the simplest level, synchronous replication means any change must be made to both sides of mirrored storage. For example, an Oracle database commits a transaction, and data is written to a redo log on synchronously mirrored storage. The storage system must not acknowledge the write operation has completed until it has been committed to nonvolatile media on both sites. Only then is it safe to proceed without the risk of data loss.

The use of synchronous replication technology is only the first step in designing and managing a synchronous replication solution. The most important consideration is to know exactly what happens during various planned and unplanned failure scenarios.

True HA data center with MetroCluster

MetroCluster replication is based on NetApp SyncMirror® technology which provides synchronous mirroring of data, implemented at the RAID level. You can use SyncMirror to create aggregates that consist of two copies of the same NetApp WAFL® file system. The two copies, known as plexes, are simultaneously updated and are always identical. This technology meets the requirements of most customers who demand synchronous replication under normal conditions.

Architecture

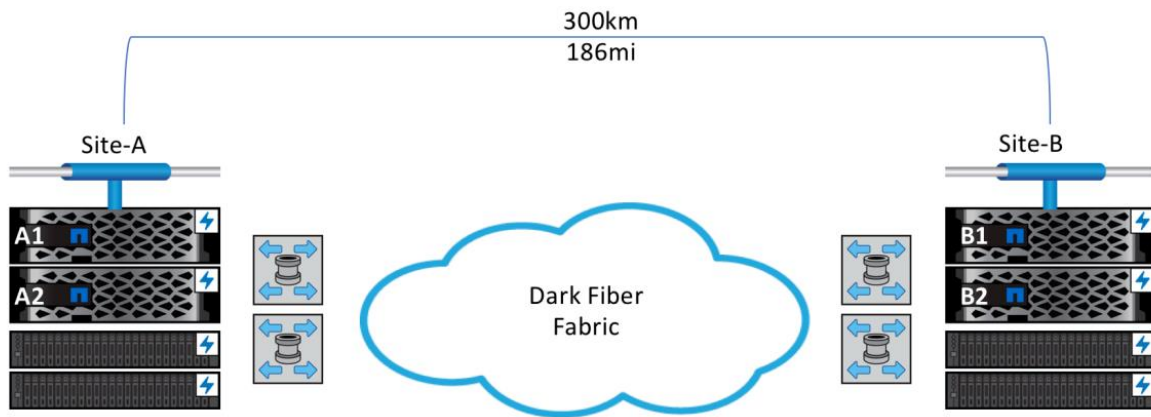
NetApp MetroCluster is designed for organizations that require continuous protection of their storage infrastructure and mission-critical business applications. By synchronously replicating data between geographically separated clusters, MetroCluster provides a zero-touch, continuously available solution that guards against faults inside and outside of the array.

MetroCluster physical architecture

MetroCluster configurations protect data by using two distinct clusters that are separated by a distance up to 700km. Each cluster synchronously mirrors the data and configuration information of the other. Effectively, all storage virtual machines (SVMs) and their associated configurations are replicated.

Independent clusters provide isolation and resilience to logical errors. If a disaster occurs at one site, an administrator can perform a switchover, which activates the mirrored SVMs and resumes serving the mirrored data from the surviving site.

The MetroCluster two-node configuration has a one-node cluster at each site. Planned and unplanned events are handled by using switchover and switchback operations. Switchover and the corresponding switchback operations transfer the entire clustered workload between the sites.

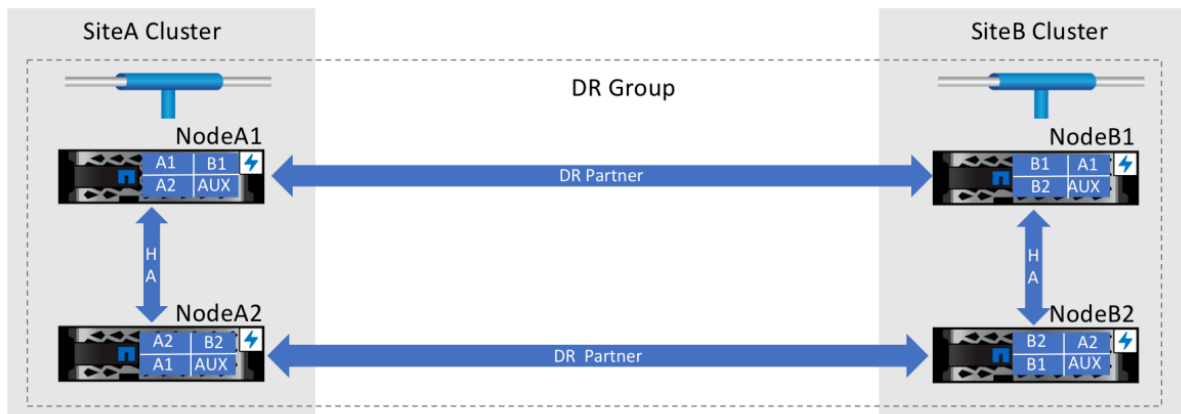


The two clusters and sites are connected by two separate networks that provide the replication transport. The cluster peering network is an IP network that is used to replicate cluster configuration information between the sites. The shared storage fabric is an FC or IP connection that is used for storage and NVRAM synchronous replication between the two clusters. For MetroCluster IP, replication uses both iWARP for NVRAM and iSCSI for disk replication. All storage is visible to all controllers through the shared storage fabric.

Function	MetroCluster FC	MetroCluster IP
MetroCluster fabric	FC ISLs	Ethernet/IP ISLs
Fabric fibre switches	Two per site	None
SAS bridges	Two per site	None
Fabric Ethernet switches	None	Two per site
FC-VI adapters	Yes Number of adapters depends on controller	None
Fabric 25G/40G/100G Ethernet adapters	None	One per node depending on platform. Adapter is used to replication both iWARP and iSCSI
Intercluster	Switchless and switched	Switchless and switched
Shelves	Physically visible to both sites	Not visible to remote clusters
NVRAM replication	FC protocol	IP/iWARP
SyncMirror replication	FC protocol	IP/iSCSI
Configuration replication services	No changes	No changes
MetroCluster size	Two, four, and eight nodes	Four and eight nodes
MetroCluster stretch	Yes	No
Advanced Disk Partitioning	No	Yes, for AFF only

Local failover (HA) and remote switchover (DR) In a two-node architecture, both HA failover and remote DR are accomplished by using MetroCluster switchover and switchback functionality. Each node acts as both the HA partner and a DR partner for its peer. NVRAM is replicated to the remote partner, like a four-node configuration.

The four-node and eight-node architectures provide both local HA failover and remote DR switchover. Each node has an HA partner in the same local cluster and a DR partner in the remote cluster, as shown in Figure 3. A1 and A2 are HA partners, as are B1 and B2. Node A1 and B1 are DR partners, as are A2 and B2. NVRAM is replicated to both the HA and the DR partner.



In a local HA failover, one of the nodes in the HA pair temporarily takes over the shared storage and services of its HA partner. For example, node A2 takes over the resources of node A1. Takeover is enabled by mirrored NVRAM and multipathed storage between the two nodes. Failover can be planned, for example, to perform a nondisruptive ONTAP upgrade, or it can be unplanned during a panic or hardware failure.

Giveback is the reverse process; the failed node resumes its resources from the node that took over. Giveback is always a planned operation. Failover is always to the local HA partner, and either node can fail over to the other. During a switchover, the peer cluster takes over the storage and services of the other cluster while still performing its own workloads. For instance, when site A switches over to site B, the nodes of cluster B temporarily assume control of the storage and services previously owned by cluster A.

Once the switchover is completed, the SVMs from cluster A are brought back online and can continue to run on cluster B. Switchover can be negotiated (planned), for example, to perform testing or site maintenance, or it can be forced (unplanned) in the event of a disaster that destroys one of the sites.

Switchback is the process in which the surviving cluster sends the switched-over resources back to their original location to restore the

steady operational state. Switchback is coordinated between the two clusters and is always a planned operation. Either site can switch over to the other.

MetroCluster replication

MetroCluster IP leverages direct attached storage, which eliminates the need for external serial-attached SCSI (SAS) bridges to connect disks to the storage fabric. Each node in the disaster recovery group acts as a storage proxy or iSCSI target that exports its disks to the other nodes in the group. iSCSI (SCSI over TCP/IP) is the storage transport protocol for the IP fabric that allows the iSCSI initiator and targets to communicate over a TCP/IP fabric. Each node in the disaster recovery group accesses its remote storage through an iSCSI initiator that establishes an iSCSI session with a remote disaster recovery partner iSCSI target.

The use of iSCSI and direct-attached storage also enables the use of systems that have internal disks. iSCSI allows the nodes to provide the disaster recovery partner node access to internal storage in addition to storage devices located in external disk shelves.

MetroCluster has three planes of replication:

1. Configuration replication
2. NVRAM replication
3. Storage replication

Configuration replication

MetroCluster configurations consists of two ONTAP clusters, each with its own replicated database (RDB) that contains its own metadata or configuration information. When a switchover occurs, the stopped

cluster's metadata objects are activated on the surviving cluster, which requires the transfer of these objects from the owning cluster to the other cluster. The transfer mechanism has three components: cluster peering, configuration replication service (CRS), and a volume that contains metadata.

- **Cluster peering** is a method of establishing a customer-supplied TCP/IP connection between two ONTAP clusters using intercluster logical interfaces (LIFs). It enables the replication of configuration objects between the clusters and is used in Metrocluster and ONTAP SnapMirror software.

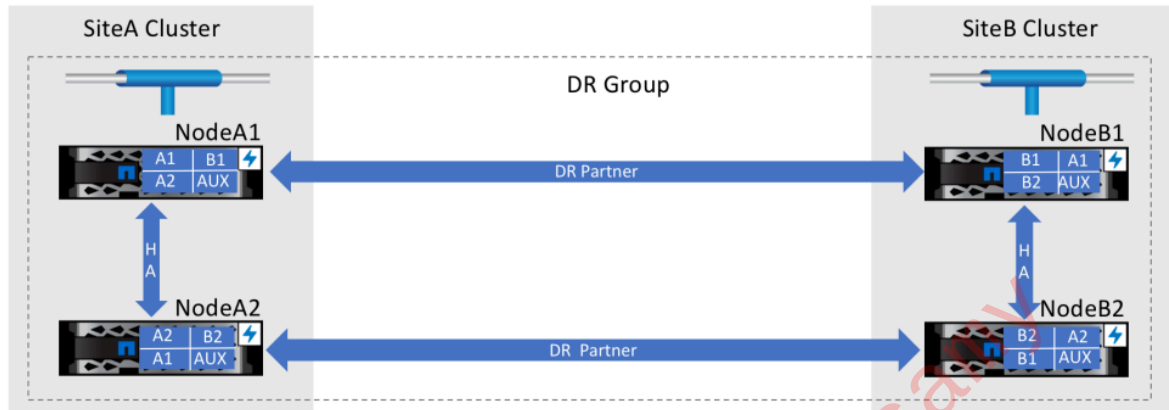
The **Configuration Replication Service** (CRS) is a component of a MetroCluster configuration that runs on each cluster and is responsible for replicating the required metadata objects from the owning cluster to the peered cluster's replicated database (RDB). This service replicates configuration objects (e.g., SVMs, LIFs, volumes, aggregates, LUNs) and protocol objects (e.g., CIFS, NFS, SAN) between the clusters using the peering network.

NVRAM replication

NVRAM replication is a process of copying the local node's NVRAM to the NVRAM of the remote disaster recovery node to protect against data loss in the event of a failover or switchover. In an ONTAP HA pair, each node mirrors its NVRAM to the other node via the HA interconnect. The NVRAM is divided into two segments, one for each node's NVRAM.

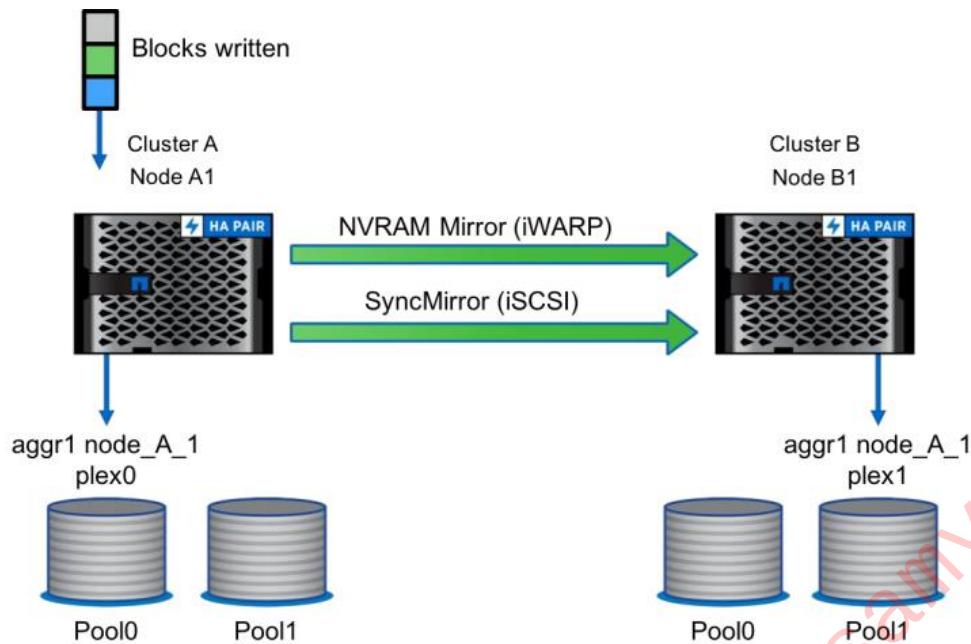
METROCLUSTER provides additional mirroring by having a DR partner node on the other site, and the NVRAM is mirrored to the DR partner via the Inter-Switch Link (ISL) connection. In a four-node configuration, each node's NVRAM is mirrored twice, once to the HA partner and

once to the DR partner, and each node's NVRAM is split into four segments.



Storage replication

Storage replication mirrors the local and remote back-end disks using RAID SyncMirror (RSM). MetroCluster IP presents the back-end storage as logically shared by making each node in a disaster recovery group serve as a remote iSCSI target. For a node to access its remote back-end disks, it goes through its remote disaster recovery partner node to access the remote disks that are served through an iSCSI target.



First check the metrocluster license.

Once all the physical connections are verified, then you can configure the metrocluster using the following command.

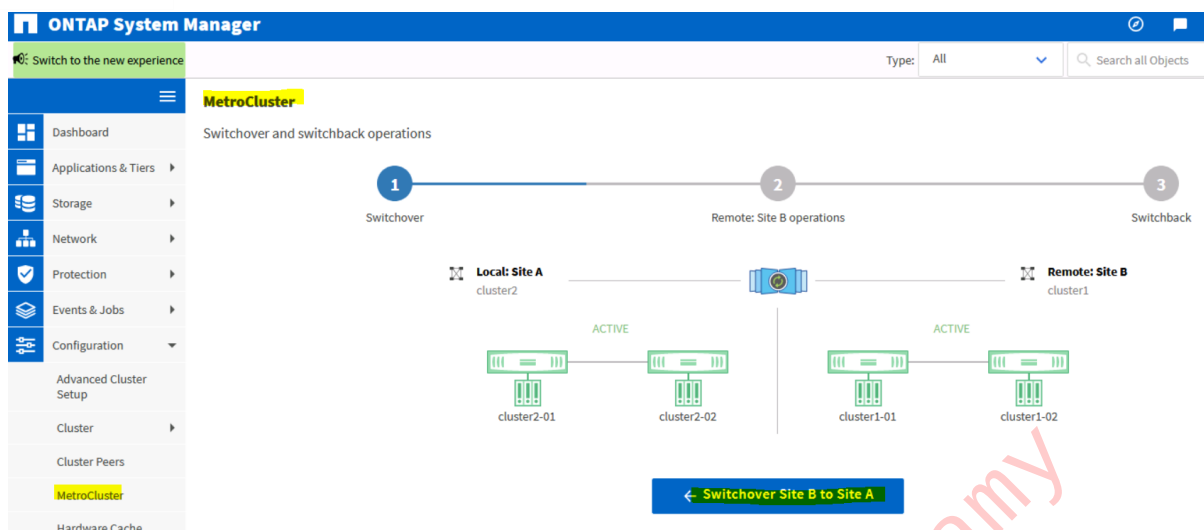
```
cluster2::> metrocluster configure
[Job 463] Job succeeded: Configure is successful.
```

```
cluster2::>
cluster2::> metrocluster node show
```

DR Group	Cluster	Node	Configuration State	DR Mirroring	Mode
1	cluster2	cluster2-01	configured	enabled	normal
		cluster2-02	configured	enabled	normal
	cluster1	cluster1-01	configured	enabled	normal
		cluster1-02	configured	enabled	normal

4 entries were displayed.

From ONTAP 9.8, you can manage the MCC using System Manager.



From CLI, use the following command to check the metrocluster status.

```
cluster2::>
cluster2::> metrocluster show

Configuration: IP-fabric

Cluster          Entry Name          State
-----
Local: cluster2
Configuration State configured
Mode               normal
AUSO Failure Domain auto-on-cluster-disaster
Remote: cluster1
Configuration State configured
Mode               normal
AUSO Failure Domain auto-on-cluster-disaster

cluster2::>
```

To list the cluster node status and details.

```
cluster2::> metrocluster node show
```

DR Group	Cluster Node	Configuration State	DR Mirroring Mode
1	cluster2		
	cluster2-01	configured	enabled normal
	cluster2-02	configured	enabled normal
	cluster1		
	cluster1-01	configured	enabled normal
	cluster1-02	configured	enabled normal

4 entries were displayed.

Check the metrocluster status, run the metrocluster check run command.

Then run, metrocluster check show command.

```
cluster2::> metrocluster check run
```

The operation has been started and is running in the background. Wait for it to complete and run "metrocluster check show" to view the results. To check the status of the running metrocluster check operation, use the command, "metrocluster operation history show -job-id 160"

```
cluster2::> metrocluster check show
```

Error: show failed: A metrocluster check operation is in progress. Wait for it to complete and retry this command. To check the status of the running metrocluster check operation, use the "metrocluster operation history show -job-id 160".

```
cluster2::>
```

```
cluster2::> metrocluster operation history show -job-id 160
```

Operation	State	Start Time	End Time
check	in-progress	7/12/2020 02:08:46	-

```
cluster2:>
cluster2:> metrocluster operation history show -job-id 160
Operation          State          Start Time      End Time
-----
check              in-progress    7/12/2020 02:08:46
-
cluster2:> metrocluster operation history show -job-id 160
Operation          State          Start Time      End Time
-----
check              successful     7/12/2020 02:08:46
                                     7/12/2020 02:10:33
- . . . -
```

Nodes, LIF's, Connections and aggregates with mirror are working fine.

```
cluster2:> metrocluster check show

Last Checked On: 7/12/2020 02:08:46

Component          Result
-----
nodes              ok
lifs               ok
config-replication ok
aggregates         ok
clusters           ok
connections        ok
volumes            ok
7 entries were displayed.
```

Check the metrocluster interconnect type (MCC-IP using iWARP).


```
cluster2::> metrocluster interconnect show
```

Node	Partner Name	Partner Type	Mirror Admin Status	Mirror Oper Status	Adapter	Type	Status
cluster2-01							
	cluster2-02	HA	enabled	online	e0f	iWARP	Up
					e0g	iWARP	Up
	cluster1-01	DR	enabled	online	e0f	iWARP	Up
					e0g	iWARP	Up
	cluster1-02	AUX	enabled	offline	e0f	iWARP	Up
					e0g	iWARP	Up
cluster2-02							
	cluster2-01	HA	enabled	online	e0f	iWARP	Up
					e0g	iWARP	Up
	cluster1-02	DR	enabled	online	e0f	iWARP	Up
					e0g	iWARP	Up
	cluster1-01	AUX	enabled	offline	e0f	iWARP	Up
					e0g	iWARP	Up

12 entries were displayed.

List the SVM details.

In MCC, SVM uses the following subtypes:

1. Sync_Source
2. Sync_destination


```
cluster2::> metrocluster vserver show
```

```
Cluster: cluster1
```

Vserver	Partner Vserver	Configuration State
cluster1	cluster2	healthy
svm1	svm1-mc	healthy

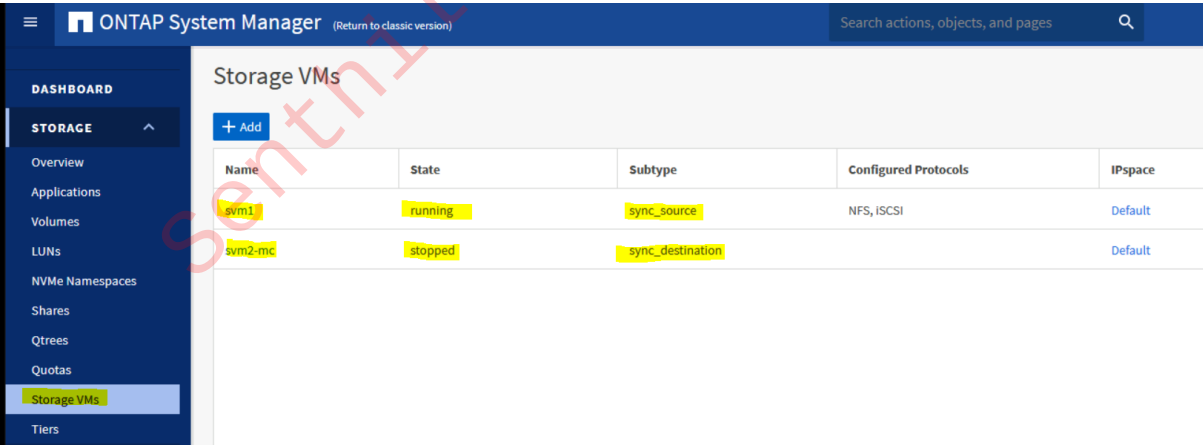
```
Cluster: cluster2
```

Vserver	Partner Vserver	Configuration State
cluster2	cluster1	healthy
svm2	svm2-mc	healthy

```
4 entries were displayed.
```

In MCC, while you are creating a SVM, automatically it creates SVM_MC with sync_destination subtype in another cluster.

As per the example, In Cluster1 svm1's subtype is Sysnc_source and svm2_mc subtype is sync_destination.



Name	State	Subtype	Configured Protocols	IPspace
svm1	running	sync_source	NFS, iSCSI	Default
svm2-mc	stopped	sync_destination		Default

In Cluster2, svm2 subtype is sync_source and svm1_mc is sync_destination.

ONTAP System Manager (Return to classic version) Search actions, objects, and pages

DASHBOARD

STORAGE

- Overview
- Applications
- Volumes
- LUNs
- NVMe Namespaces
- Shares
- Qtrees
- Quotas
- Storage VMs**
- Tiers

Storage VMs

[+ Add](#)

Name	State	Subtype	Configured Protocols	IPspace
svm1-mc	stopped	sync_destination		Default
svm2	running	sync_source	NFS, iSCSI	Default

```
cluster2::> vserver create -vserver svm_new -rootvolume svm_new_root -rootvolume-security-style
unix -language C.UTF-8 -snapshot-policy default -data-services data-iscsi,data-nfs,data-cifs,dat
a-flexcache -foreground true
[Job 469] Job succeeded:
[Job 469] Job succeeded:
Vserver creation completed.
```

```
cluster2::> vserver show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster2	admin	-	-	-	-	-
cluster2-01	node	-	-	-	-	-
cluster2-02	node	-	-	-	-	-
svm1-mc	data	sync-destination	running	stopped	svm1_root	aggr1_cluster1_01
svm2	data	sync-source	running	running	svm2_root	aggr1_cluster2_01
svm_new	data	sync-source	running	running	svm_new_root	aggr1_cluster2_02

6 entries were displayed.

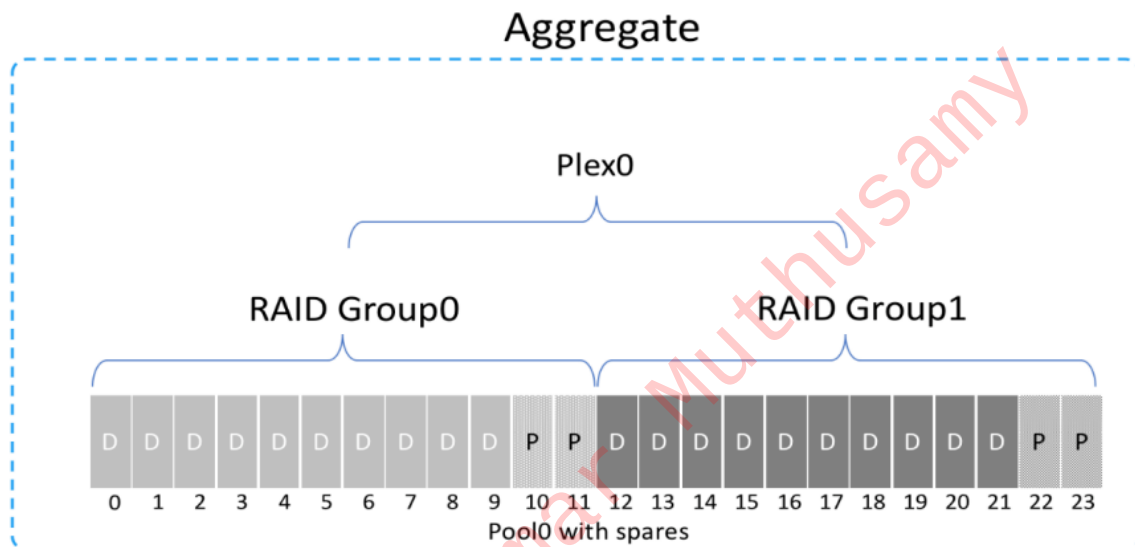
```
cluster1::> vserver show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster1	admin	-	-	-	-	-
cluster1-01	node	-	-	-	-	-
cluster1-02	node	-	-	-	-	-
svm1	data	sync-source	running	running	svm1_root	aggr1_cluster1_01
svm2-mc	data	sync-destination	running	stopped	svm2_root	aggr1_cluster2_01
svm_new-mc	data	sync-destination	running	stopped	svm_new_root	aggr1_cluster2_02

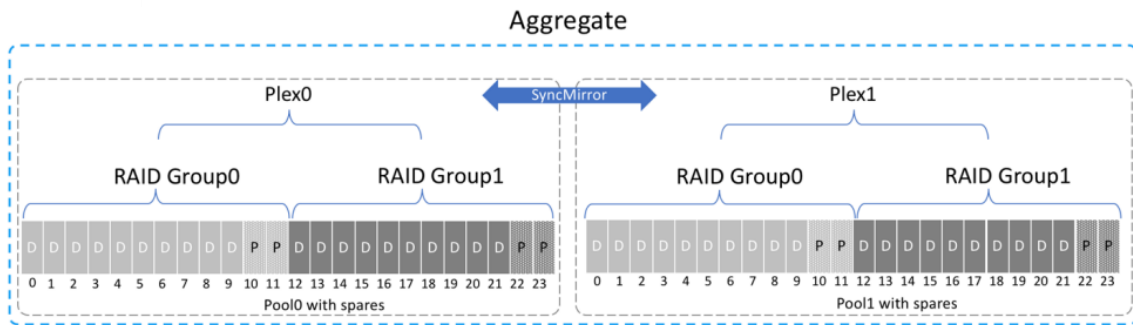
6 entries were displayed.

SyncMirror storage replication

An ONTAP system stores data in NetApp FlexVol® volumes that are provisioned from aggregates. Each aggregate contains a WAFL file system. In a configuration without MetroCluster, the disks in each aggregate consist of a single or multiple RAID group, known as a plex. The plex resides in local storage attached to the controller.



In a MetroCluster configuration, each aggregate consists of two plexes that are physically separated: a local plex and a remote plex. All storage is shared and is visible to all the controllers in the MetroCluster configuration. The local plex must contain only disks from the local pool (pool0), and the remote plex must contain only disks from the remote pool. The local plex is always plex0. Each remote plex has a number other than 0 to indicate that it is remote (for example, plex1 or plex2).



List the aggregates. All the aggregates are mirrored.

```
cluster2::> aggr show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0_cluster2_01	9.10GB	434.7MB	95%	online	1	cluster2-01	raid_dp, mirrored, normal
aggr0_cluster2_02	9.10GB	466.5MB	95%	online	1	cluster2-02	raid_dp, mirrored, normal
aggr1_cluster2_01	57.69GB	57.03GB	1%	online	4	cluster2-01	raid_dp, mirrored, normal
aggr1_cluster2_02	72.89GB	72.87GB	0%	online	1	cluster2-02	raid_dp, mirrored, normal

4 entries were displayed.

In cluster2, create new aggregate using (`-mirror true`) option. It creates two plexes.

```
cluster2::> aggr create -aggregate aggr_new -diskcount 4 -raidtype raid4 -mirror true -node cluster2-01
```

Info: The layout for aggregate "aggr_new" on node "cluster2-01" would be:

First Plex

RAID Group rg0, 2 disks (block checksum, raid4)

Position	Disk	Type	Usable Size	Physical Size
parity	VMw-1.69	VMDISK	-	-
data	VMw-1.38	VMDISK	28.42GB	28.44GB

Second Plex

RAID Group rg0, 2 disks (block checksum, raid4)

Position	Disk	Type	Usable Size	Physical Size
parity	0m.i1.0L55	VMDISK	-	-
data	0m.i1.0L25	VMDISK	28.42GB	28.44GB

Aggregate capacity available for volume use would be 24.30GB.

```
Do you want to continue? {y|n}: y
[Job 468] Job succeeded: DONE
```

```
cluster2::> aggr show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0_cluster2_01	9.10GB	452.7MB	95%	online	1	cluster2-01	raid_dp, mirrored, normal
aggr0_cluster2_02	9.10GB	452.7MB	95%	online	1	cluster2-02	raid_dp, mirrored, normal
aggr1_cluster2_01	57.69GB	57.14GB	1%	online	4	cluster2-01	raid_dp, mirrored, normal
aggr1_cluster2_02	72.89GB	72.87GB	0%	online	1	cluster2-02	raid_dp, mirrored, normal
aggr_new	24.30GB	24.30GB	0%	online	0	cluster2-01	raid4, mirrored, normal

5 entries were displayed.


```
cluster1::> aggr plex offline -aggregate aggr1_cluster1_02 -plex plex0
```

```
Plex offline successful on plex: aggr1_cluster1_02/plex0
```

```
cluster1::> aggr plex show -aggregate aggr1_cluster1_02
```

Aggregate	Plex	Is Online	Is Resyncing	Resyncing Percent	Status
aggr1_cluster1_02	plex0	false	false		- normal, inactive
aggr1_cluster1_02	plex1	true	false		- normal, active

2 entries were displayed.

```
cluster1::> metrocluster check show
```

```
Last Checked On: 4/26/2019 02:52:48
```

Component	Result
nodes	ok
lifs	ok
config-replication	ok
aggregates	warning
clusters	ok
connections	not-applicable

6 entries were displayed.

```
cluster1::> aggr plex online -aggregate aggr1_cluster1_02 -plex plex0
```

```
Plex online successful on plex: aggr1_cluster1_02/plex0
```

```
cluster1::>
```

```
cluster1::> aggr plex show -aggregate aggr1_cluster1_02
```

Aggregate	Plex	Is Online	Is Resyncing	Resyncing Percent	Status
aggr1_cluster1_02	plex0	true	false		- normal, active
aggr1_cluster1_02	plex1	true	false		- normal, active

2 entries were displayed.

```
cluster1:> metrocluster check run
```

The operation has been started and is running in the background. Wait for it to complete and run "metrocluster check show" to view the results. To check the status of the running metrocluster check operation, use the command, "metrocluster operation history show -job-id 313"

```
cluster1:> metrocluster check show
```

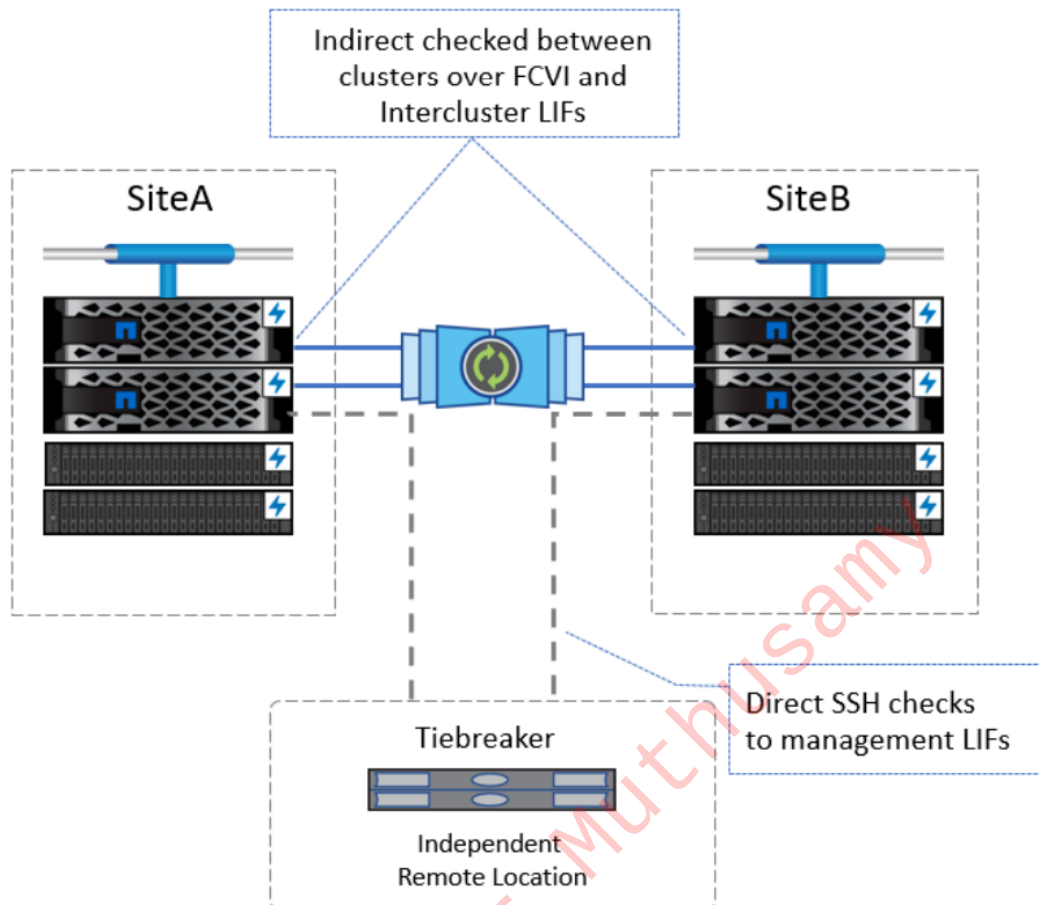
Last Checked On: 4/26/2019 02:55:44

Component	Result
nodes	ok
lifs	ok
config-replication	ok
aggregates	ok
clusters	ok
connections	not-applicable

6 entries were displayed.

NetApp Tiebreaker

The MetroCluster Tiebreaker software alerts you if all connectivity between the sites is lost. The MetroCluster Tiebreaker software supports all the MetroCluster configurations that are supported in ONTAP 8.3 and 9.0 through 9.7. The Tiebreaker software resides on a Linux host. You need Tiebreaker software only if you want to monitor two clusters and the connectivity status between them from a third site. Doing so enables each partner in a cluster to distinguish between an ISL failure, when intersite links are down, from a site failure.



Detecting failures with MetroCluster Tiebreaker

The Tiebreaker software resides on a Linux host. You need the Tiebreaker software only if you want to monitor two clusters and the connectivity status between them from a third site. Doing so enables each partner in a cluster to distinguish between an ISL failure, when intersite links are down, from a site failure. After you install the Tiebreaker software on a Linux host, you can configure the clusters in a MetroCluster configuration to monitor for disaster conditions.

Add ONTAP tie-breaker host to metrocluster.


```
cluster2:> metrocluster configuration-settings mediator add -mediator-address 192.168.0.61
Adding the mediator and enabling Automatic Unplanned Switchover. It may take a few minutes to complete.
Please enter the username for the mediator: mediatoradmin
Please enter the password for the mediator:
Confirm the mediator password:
Successfully added the mediator.
```

In Tie-Breaker installed linux host, check the HA status using the following command (monitor show –status)

1. Monitor State - Normal

```
[root@tiebreak ~]# netapp-metrocluster-tiebreaker-software-cli
NetApp MetroCluster Tiebreaker :>
NetApp MetroCluster Tiebreaker :>
NetApp MetroCluster Tiebreaker :> monitor show -status
MetroCluster: MCC-lab
  Disaster: false
  Monitor State: Normal
  Observer Mode: true
  Silent Period: 10
  Override Vetoes: false
  Cluster: cluster1 (UUID:8c008c78-a89f-11e8-902d-005056b0964c)
    Reachable: true
    Intersite Connectivity Available: true
      Node: cluster1-01 (UUID:62ef9e53-a89e-11e8-902d-005056b0964c)
        Reachable: true
        Intersite Connectivity Available: true
        State: normal
      Node: cluster1-02 (UUID:6560d407-a89e-11e8-852b-005056b04a4c)
        Reachable: true
        Intersite Connectivity Available: true
        State: normal
  Cluster: cluster2 (UUID:cc88c6b1-a89f-11e8-89ca-005056b03e9a)
    Reachable: true
    Intersite Connectivity Available: true
      Node: cluster2-01 (UUID:68b5e996-a89e-11e8-89ca-005056b03e9a)
        Reachable: true
        Intersite Connectivity Available: true
        State: normal
      Node: cluster2-02 (UUID:6b3dcd3a-a89e-11e8-baf8-005056b03440)
        Reachable: true
        Intersite Connectivity Available: true
        State: normal
```

Performing switchover for tests or maintenance

If you want to test the MetroCluster functionality or to perform planned maintenance, you can perform a negotiated switchover in which one cluster is cleanly switched over to the partner cluster. You can then heal and switch back the configuration.

Creating a new SVM (vs_new).

Cluster1 – vs_new - running state

cluster2 – vs_new_mc --- Stopped state.

```
cluster1::> vservers create -vservers vs_new -rootvolume svm_root -rootvolume-security-style unix -language C.UTF-8 -snapshot-policy default -foreground true -aggregate aggr1_cluster1_01
[Job 389] Job succeeded:
Vserver creation completed.
```

```
cluster1::> vservers show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster1	admin	-	-	-	-	-
cluster1-01	node	-	-	-	-	-
cluster1-02	node	-	-	-	-	-
svm1	data	sync-source	running	running	svm1_root	aggr1_cluster1_02
svm2-mc	data	sync-destination	running	stopped	svm2_root	aggr1_cluster2_01
vs_new	data	sync-source	running	running	svm_root	aggr1_cluster1_01

6 entries were displayed.

```
cluster2::> vservers show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster2	admin	-	-	-	-	-
cluster2-01	node	-	-	-	-	-
cluster2-02	node	-	-	-	-	-
svm1-mc	data	sync-destination	running	stopped	svm1_root	aggr1_cluster1_02
svm2	data	sync-source	running	running	svm2_root	aggr1_cluster2_01
vs_new-mc	data	sync-destination	running	stopped	svm_root	aggr1_cluster1_01

6 entries were displayed.

In cluster2 , create new SVM (vs_dr).

```
cluster2::> vservers create -vservers vs_dr -rootvolume svm_root -rootvolume-security-style unix -language C.UTF-8 -snapshot-policy default -foreground true -aggregate aggr1_cluster2_01
[Job 123] Job succeeded:
Vserver creation completed.
```

```
cluster2::> vservers show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster2	admin	-	-	-	-	-
cluster2-01	node	-	-	-	-	-
cluster2-02	node	-	-	-	-	-
svm1-mc	data	sync-destination	running	stopped	svm1_root	aggr1_cluster1_02
svm2	data	sync-source	running	running	svm2_root	aggr1_cluster2_01
vs_dr	data	sync-source	running	running	svm_root	aggr1_cluster2_01
vs_new-mc	data	sync-destination	running	stopped	svm_root	aggr1_cluster1_01

```
cluster1::> vservers show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
cluster1	admin	-	-	-	-	-
cluster1-01	node	-	-	-	-	-
cluster1-02	node	-	-	-	-	-
svm1	data	sync-source	running	running	svm1_root	aggr1_cluster1_02
svm2-mc	data	sync-destination	running	stopped	svm2_root	aggr1_cluster2_01
vs_dr-mc	data	sync-destination	running	stopped	svm_root	aggr1_cluster2_01
vs_new	data	sync-source	running	running	svm_root	aggr1_cluster1_01

7 entries were displayed.

Create vservers peer between these SVM's
(vsnew and vs_dr)

```
cluster1::> vserver peer create -vserver vs_new -peer-vserver vs_dr -application
s snapmirror -peer-cluster cluster2

Info: [Job 391] 'vserver peer create' job queued

cluster1::> vserver peer show
```

Vserver	Peer Vserver	Peer State	Peer Cluster	Peering Applications	Remote Vserver
vs_new	vs_dr	initializing	cluster2	snapmirror	vs_dr

```
cluster2::> vserver peer show
```

Vserver	Peer Vserver	Peer State	Peer Cluster	Peering Applications	Remote Vserver
vs_dr	vs_new	pending	cluster1	snapmirror	vs_new

```
cluster2::> vserver peer accept -vserver vs_dr -peer-vserver vs_new

Info: [Job 124] 'vserver peer accept' job queued

cluster2::> vserver peer show
```

Vserver	Peer Vserver	Peer State	Peer Cluster	Peering Applications	Remote Vserver
vs_dr	vs_new	peered	cluster1	snapmirror	vs_new

Assume disaster strikes, so you can't access the cluster1 (regional outage) - Unplanned

```
cluster2::> metrocluster show
```

Configuration: IP-fabric

Cluster	Entry Name	State
Local: cluster2	Configuration State	configured
	Mode	switchover
	AUSO Failure Domain	auso-on-cluster-disaster
	Configuration State	not-reachable
Remote: cluster1	Mode	-
	AUSO Failure Domain	not-reachable

Check in Tie-Breaker (Quorum Witness)

State: Unable to reach cluster1.

```
NetApp MetroCluster Tiebreaker :> monitor show -status
MetroCluster: MCC-lab
  Disaster: false
  Monitor State: MCCTB is unable to reach cluster "cluster1"
  Observer Mode: true
  Silent Period: 10
  Override Vetoes: false
  Cluster: cluster1 (UUID:8c008c78-a89f-11e8-902d-005056b0964c)
    Reachable: false
    Intersite Connectivity Available: unknown
      Node: cluster1-01 (UUID:62ef9e53-a89e-11e8-902d-005056b0964c)
        Reachable: false
        Intersite Connectivity Available: unknown
        State: normal
      Node: cluster1-02 (UUID:6560d407-a89e-11e8-852b-005056b04a4c)
        Reachable: false
        Intersite Connectivity Available: unknown
        State: normal
  Cluster: cluster2 (UUID:cc88c6b1-a89f-11e8-89ca-005056b03e9a)
    Reachable: true
    Intersite Connectivity Available: false
      Node: cluster2-01 (UUID:68b5e996-a89e-11e8-89ca-005056b03e9a)
        Reachable: true
        Intersite Connectivity Available: false
        State: normal
      Node: cluster2-02 (UUID:6b3dcd3a-a89e-11e8-baf8-005056b03440)
        Reachable: true
        Intersite Connectivity Available: false
        State: normal
```

It confirms, disaster strikes.

```
NetApp MetroCluster Tiebreaker :> monitor show -status
MetroCluster: MCC-lab
Disaster: true
Monitor State: Disaster confirmed.
Observer Mode: true
Silent Period: 10
Override Vetoes: false
Cluster: cluster1 (UUID: 8c008c78-a89f-11e8-902d-005056b0964c)
  Reachable: false
  Intersite Connectivity Available: unknown
    Node: cluster1-01 (UUID: 62ef9e53-a89e-11e8-902d-005056b0964c)
      Reachable: false
      Intersite Connectivity Available: unknown
      State: normal
    Node: cluster1-02 (UUID: 6560d407-a89e-11e8-852b-005056b04a4c)
      Reachable: false
      Intersite Connectivity Available: unknown
      State: normal
Cluster: cluster2 (UUID: cc88c6b1-a89f-11e8-89ca-005056b03e9a)
  Reachable: true
  Intersite Connectivity Available: false
    Node: cluster2-01 (UUID: 68b5e996-a89e-11e8-89ca-005056b03e9a)
      Reachable: true
      Intersite Connectivity Available: false
      State: normal
    Node: cluster2-02 (UUID: 6b3dcd3a-a89e-11e8-baf8-005056b03440)
      Reachable: true
      Intersite Connectivity Available: false
      State: normal
```

Now you can perform the switchover from cluster2.

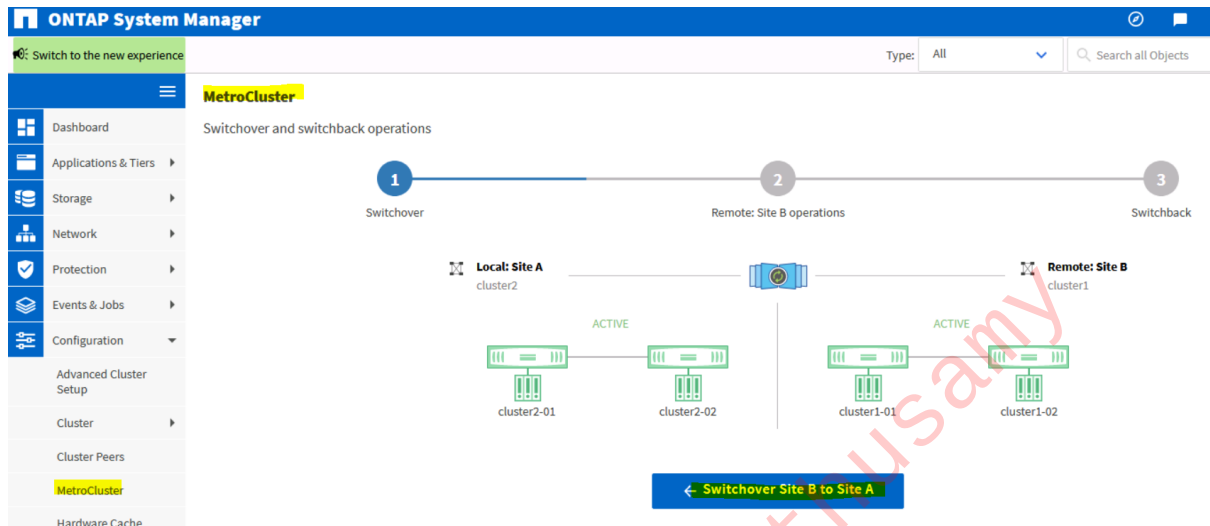
(metrocluster switchover -forced-on-disaster)

```
cluster2::*> metrocluster switchover -forced-on-disaster

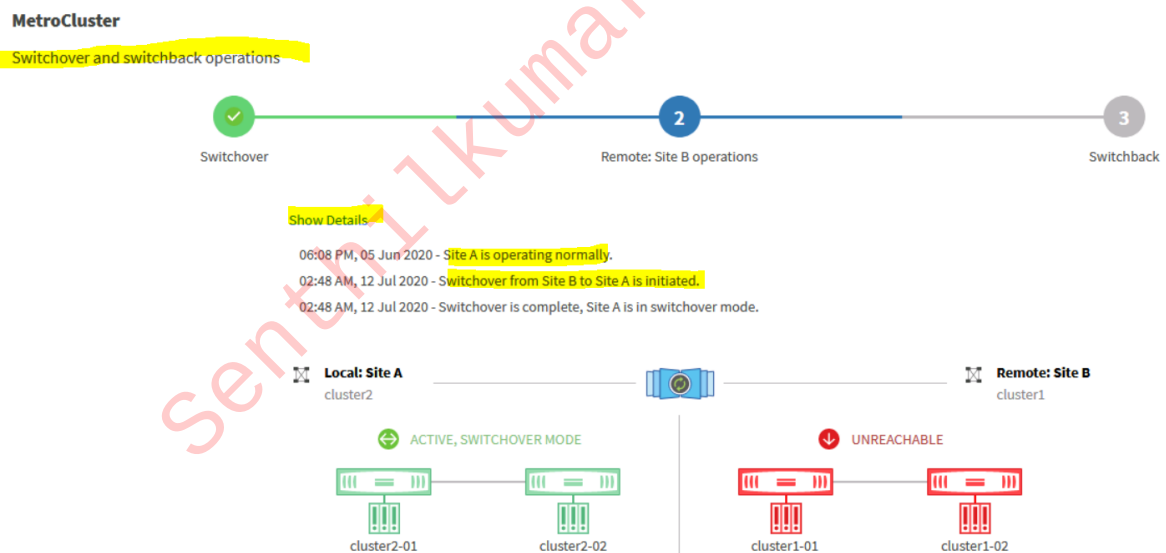
Warning: MetroCluster switchover is a disaster recovery operation that could
        cause some data loss. The nodes on the other site must either be
        prevented from serving data or be simply powered off.
Do you want to continue? {y|n}: y
[Job 157] Synchronizing cluster configuration from the remote cluster (attempt 1
[Job 157] Job succeeded: Switchover is successful.
```

Same operation, you can do it in using System Manager also.

Normal state – both are working.



Unplanned Failover – Disaster Strikes.



Check the switch over status using the following command.

(metrocluster node show)

Cluster1 – unreachable.

Switchover completed successfully.

```
cluster2::> metrocluster node show
```

DR Group	Cluster	Node	Configuration State	DR Mirroring Mode
1	cluster2	cluster2-01	configured	enabled switchover completed
		cluster2-02	configured	enabled switchover completed
	cluster1	cluster1-01	unreachable	- switched over
		cluster1-02	unreachable	- switched over

4 entries were displayed.

Now the SVM, vs_new_mc (subtype=sync_destination) is running state.

```
cluster2::*> vservers show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
Cluster	system	-	-	-	-	-
cluster2	admin	-	-	-	-	-
cluster2-01	node	-	-	-	-	-
cluster2-02	node	-	-	-	-	-
svm1-mc	data	sync-destination	running	running	svm1_root	aggr1_cluster1_02
svm2	data	sync-source	running	running	svm2_root	aggr1_cluster2_01
vs_dr	data	sync-source	running	running	svm_root	aggr1_cluster2_01
vs_new-mc	data	sync-destination	running	running	svm_root	aggr1_cluster1_01

8 entries were displayed.

Run to check the last metrocluster operation.

(metrocluster operation show)

Switchover operation successful.

```
cluster2::> metrocluster operation
             history show

cluster2::> metrocluster operation show
             Operation: switchover
             State: successful
             Start Time: 7/12/2020 02:48:31
             End Time: 7/12/2020 02:48:56
             Errors: -
```

Aggregates are working, but degraded state.

```
cluster2::> storage aggr show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0_cluster1_01	-	-	-	unknown	-	cluster2-01	-
aggr0_cluster1_02	-	-	-	unknown	-	cluster2-02	-

```
cluster1 Switched Over Aggregates:
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr1_cluster1_01	61.38GB	54.30GB	12%	online	4	cluster2-01	raid_dp, mirror degraded
aggr1_cluster1_02	27.29GB	27.27GB	0%	online	1	cluster2-02	raid_dp, mirror degraded

```
cluster2 Aggregates:
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0_cluster2_01	9.10GB	452.7MB	95%	online	1	cluster2-01	raid_dp, mirror

Switch Back:

After you heal the MetroCluster configuration, you can perform the MetroCluster switchback operation. The MetroCluster switchback operation returns the configuration to its normal operating state, with the sync-source storage virtual machines (SVMs) on the disaster site active and serving data from the local disk pools.

- The disaster cluster must have successfully switched over to the surviving cluster.
- Healing must have been performed on the data and root aggregates.
- The surviving cluster nodes must not be in the HA failover state (all nodes must be up and running for each HA pair).
- The disaster site controller modules must be completely booted and not in the HA takeover mode.
- The root aggregate must be mirrored.
- The Inter-Switch Links (ISLs) must be online.

Before switchback, you must heal the data and root aggregates using the following command.

(metrocluster heal -phase aggregates) – for data aggregates.

```
cluster2::*> metrocluster heal -phase aggregates
[Job 158] Job succeeded: Heal Aggregates is successful.

cluster2::*> metrocluster heal -phase root-aggregates
[Job 159] Job succeeded: Heal Root Aggregates is successful.

cluster2::*> metrocluster operation show
  Operation: heal-root-aggregates
    State: successful
  Start Time: 4/26/2019 03:46:26
    End Time: 4/26/2019 03:46:27
    Errors: -
```

Check the metrocluster node, root aggregates are healed successfully.

```
cluster2::*> metrocluster node show
DR
Group Cluster Node Configuration State DR Mirroring Mode
-----
1 cluster2
    cluster2-01 configured enabled heal roots completed
    cluster2-02 configured enabled heal roots completed
  cluster1
    cluster1-01 unreachable - switched over
    cluster1-02 unreachable - switched over
4 entries were displayed.
```

Verifying that your system is ready for a switchover

You can use the `-simulate` option to preview the results of a switchover operation. A verification check gives you a way to verify that most of the preconditions for a successful run are met before you start the operation. Issue these commands from the site that will remain up and operational:

Run the following command to switchback.

(metrocluster switchback –simulate)

```
cluster2::*> cluster peer show
Peer Cluster Name Cluster Serial Number Availability Authentication
-----
cluster1 1-80-000011 Unavailable ok

cluster2::*> metrocluster switchback -simulate
[Job 160] Synchronizing cluster configuration from the local cluster to the remo
[Job 160] Synchronizing cluster configuration from the local cluster to the remo
[Job 160] Synchronizing cluster configuration from the local cluster to the remo
[Job 160] Job succeeded: Switchback simulation is successful.
```

Once the simulation is successful, perform the switchback operation.

```
cluster2::*> metrocluster switchback

Warning: switchback is about to start. It will stop all the switched over data
Vservers on cluster "cluster2" and automatically restart them on
cluster "cluster1".
Do you want to continue? {y|n}: y
[Job 161] Synchronizing cluster configuration from the local cluster to the remo
[Job 161] Synchronizing cluster configuration from the local cluster to the remo
[Job 161] Queueing the "MetroCluster Switchback Continuation Agent" job on the r
[Job 161] Waiting for the "MetroCluster Switchback Continuation Agent" job to co
[Job 161] Job succeeded: Switchback is successful.
```

After successful switchback, check the metrocluster status.

state:normal.

```
cluster2::*> metrocluster show
Configuration: stretch

Cluster          Entry Name          State
-----
Local: cluster2
                  Configuration State configured
                  Mode normal
                  AUSO Failure Domain auso-on-cluster-disaster
Remote: cluster1
                  Configuration State configured
                  Mode normal
                  AUSO Failure Domain auso-on-cluster-disaster
```

Check the metrocluster last operation state.

Switchback successfully completed.

```
cluster1::> metrocluster show
Configuration: stretch

Cluster          Entry Name          State
-----
Local: cluster1
                  Configuration State configured
                  Mode normal
                  AUSO Failure Domain auso-on-cluster-disaster
Remote: cluster2
                  Configuration State configured
                  Mode normal
                  AUSO Failure Domain auso-on-cluster-disaster

cluster1::>
cluster1::> metrocluster operation show
  Operation: switchback-continuation-agent
  State: successful
  Start Time: 4/26/2019 04:00:47
  End Time: 4/26/2019 04:01:03
  Errors: -
```


You can perform the switchback operation using ONTAP System Manager also.

1. Heals data and root aggregates.

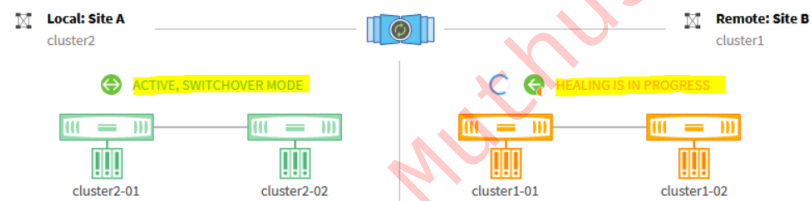
MetroCluster

Switchover and switchback operations



Show Details

06:08 PM, 05 Jun 2020 - Site A is operating normally.
 02:48 AM, 12 Jul 2020 - Switchover from Site B to Site A is initiated.
 02:48 AM, 12 Jul 2020 - Switchover is complete, Site A is in switchover mode.
 03:02 AM, 12 Jul 2020 - Site B healing is in progress.



2. Site is up and ready for switchback.

MetroCluster

Switchover and switchback operations



Show Details

06:08 PM, 05 Jun 2020 - Site A is operating normally.
 02:48 AM, 12 Jul 2020 - Switchover from Site B to Site A is initiated.
 02:48 AM, 12 Jul 2020 - Switchover is complete, Site A is in switchover mode.
 03:02 AM, 12 Jul 2020 - Site B healing is in progress.
 03:03 AM, 12 Jul 2020 - Site B healing has completed and ready for switchback.



Switchback from Site A to Site B →

Warning

You are about to switchback Site B's MetroCluster control and data services from Site A to Site B.

This operation might take a while to complete.

Are you sure you want to continue?

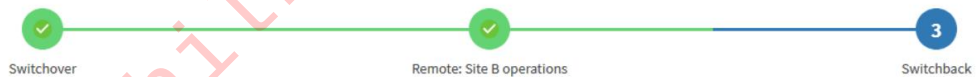
Cancel

Yes

02:48 AM, 12 Jul 2020 - Switchover is complete. Site A is in switchover mode.

MetroCluster

Switchover and switchback operations



[Show Details](#)

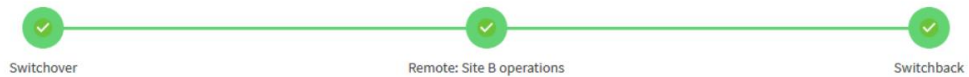
06:08 PM, 05 Jun 2020 - Site A is operating normally.
 02:48 AM, 12 Jul 2020 - Switchover from Site B to Site A is initiated.
 02:48 AM, 12 Jul 2020 - Switchover is complete, Site A is in switchover mode.
 03:02 AM, 12 Jul 2020 - Site B healing is in progress.
 03:03 AM, 12 Jul 2020 - Site B healing has completed and ready for switchback.



Switchback is successful, now both the sites are active.

MetroCluster

Switchover and switchback operations



Show Details

06:08 PM, 05 Jun 2020 - Site A is operating normally.
02:48 AM, 12 Jul 2020 - Switchover from Site B to Site A is initiated.
02:48 AM, 12 Jul 2020 - Switchover is complete, Site A is in switchover mode.
03:02 AM, 12 Jul 2020 - Site B healing is in progress.
03:03 AM, 12 Jul 2020 - Site B healing has completed.
03:05 AM, 12 Jul 2020 - Switchback from Site B to Site A is initiated.
03:06 AM, 12 Jul 2020 - Switchback has completed, and both Site A and Site B are operating normally.

